



Technical University
of Denmark

02450 INTRODUCTION TO MACHINE LEARNING AND DATA MINING

Assignment 1

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March 6th, 2025

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1 Contributions

All participants in the assignment have compiled and/or reviewed the present document and all the source codes. Nevertheless, for organizational reasons, we decided to be responsible for different parts of the document, as shown in the following (*not exhaustive*) table:

	Flavio Sarno (s242991)	Francesco Balducci (s250200)	Florencia Illanes (s247222)
Introduction	30%	30%	40%
Dataset description	40%	40%	20%
Data visualization	30%	45%	25%
PCA	50%	30%	20%
Discussion	20%	20%	60%

Table 1: Table of responsibilities

2 Introduction

The data set [1] features several [engine characteristics](#) and specifications of cars built between 1970 and 1982. The data set investigates city-cycle fuel consumption for cars built in three different geographical areas: the United States, Europe, and Japan.

The available properties are: mpg, cylinders, displacement, horsepower, weight, acceleration, model year, origin, and car name.

Generally speaking, the problem we would like to solve is to predict or compute relationships between these characteristics, mainly fuel efficiency for cars with different attributes. We will leverage machine learning techniques to tackle this task.

2.1 Literature review

This dataset was previously used in the American Statistical Association (ASA) Exposition in 1983. The data was collected by Ernesto Ramos and David Donoho, initially containing a total of 406 observations along with the same variables present in the current version [2]. The dataset provided by Ross Quinlan has 8 fewer observations due to missing values in the original database. Quinlan used this data to identify patterns in fuel consumption by combining instance-based and rule-based models. Additionally, Quinlan (1993) explored different machine learning approaches, including decision trees, linear regression, and neural networks, to improve fuel consumption prediction. His research demonstrated that combining instance-based learning with model-based methods, such as model trees, improved predictive accuracy compared to using each method separately. [3]

2.2 Goals

The [features](#) have been analyzed through Machine Learning techniques with the objective of (1) identifying and representing the relations or trends between them, (2) reducing the size of the data set while preserving the core information, (3) modeling a classification engine to predict the *origin* feature based on the rest of the data, and (4) another model to perform regression on the feature miles per gallon (*mpg*) feature, the vehicle's fuel efficiency, based on all other characteristics.

2.2.1 Classification

We would like to classify existing and unknown observations around the *origin* feature through the remaining noncategorical properties (*mpg*, *cylinders*, *displacement*, *horsepower*, *weight*, *acceleration*, *model year*). Doing so will allow the user to obtain useful information about the relationship that exists between the characteristics of a car and the expected area of utilization.

2.2.2 Regression

The main objective of this analysis is to predict the fuel efficiency of a car, measured in miles per gallon (*mpg*), based on various technical characteristics of the car. The selected predictive features are *number of cylinders*, *displacement*, *weight*, *horsepower*, *acceleration*, and *model year*. In simple terms, the goal is to model fuel consumption based on those key vehicle specifications.

3 Dataset attributes

Feature	Description	Attribute	Type
mpg	Miles per gallon (fuel efficiency)	Continuous	Ratio
cylinders	Number of engine cylinders	Discrete	Ordinal
displacement	Engine displacement (cubic inches)	Continuous	Ratio
horsepower	Engine power output	Continuous	Ratio
weight	Vehicle weight (lbs)	Continuous	Ratio
acceleration	Time to accelerate from 0 to 60 mph (seconds)	Continuous	Ratio
model_year	Model year of the vehicle	Discrete	Interval
origin	Country of origin (1: USA, 2: Europe, 3: Japan)	Discrete	Nominal
car_name	Name of the vehicle model	Discrete	Categorical

Table 2: Description and classification of dataset features based on attribute type. This dataset uses the imperial system of units [1].

The following table shows a summary of the dataset values based on the different ratio attributes, as measured by the most common statistics:

Statistic	mpg	displacement	horsepower	weight	acceleration
Count	392.00	392.00	392.00	392.00	392.00
Mean	23.45	194.41	104.47	2977.58	15.54
Std	7.81	104.64	38.49	849.40	2.76
Min	9.00	68.00	46.00	1613.00	8.00
25%	17.00	105.00	75.00	2225.25	13.78
50%	22.75	151.00	93.50	2803.50	15.50
75%	29.00	275.75	126.00	3614.75	17.03
Max	46.60	455.00	230.00	5140.00	24.80

Table 3: Summary statistics of the ratio features in the dataset.

As widely known, these metrics should not be considered as the only reliable (if any) representation of the data, but rather as a reference. Please refer to the [data visualization](#) section for in depth data illustration.

3.1 Observations tuning

The following sections will review modifications made to the dataset in order to improve its performance for the intended [goals](#).

3.1.1 Missing values

Given that the dataset consists of **398** observations, and considering that there are only **6** observation with missing properties (all on the *horsepower* attribute), they will be excluded from the analysis in order to maintain data integrity.

3.1.2 Standardization

The various attributes happen to have very different scaling: for example, *weight* is continuous 1613.0÷5140.0 while *cylinders* is 3÷8 and discrete.

Therefore, we decided to apply feature standardization to the numerical ones (all except *car_name*) ensuring that the different magnitudes would not influence the performance of the algorithms, such as Principal Component Analysis (PCA). Standardization rescales the data such that each feature has a mean of 0 and a standard deviation of 1 and was applied through the following formula:

$$z_{ij} = \frac{x_{ij} - \mu_i}{\sigma_i} \quad (1)$$

Where:

- z_{ij} is the standardized value of feature i for observation j ,
- x_{ij} is the original value of feature i for observation j ,
- μ_i is the mean of feature i across all observations,
- σ_i is the standard deviation of feature i across all observations.

3.1.3 Correlation

The following table shows how related the attributes in the dataset are. By displaying correlation coefficients, it helps identify which features are strongly or weakly related to the target variable, allowing for better feature selection in machine learning models.

	mpg	cylinders	displacement	horsepower	weight	acceleration
mpg	1.0000	-0.7776	-0.8051	-0.7784	-0.8322	0.4233
cylinders	-0.7776	1.0000	0.9508	0.8429	0.8975	-0.5047
displacement	-0.8051	0.9508	1.0000	0.8973	0.9330	-0.5438
horsepower	-0.7784	0.8430	0.8973	1.0000	0.8645	-0.6892
weight	-0.8322	0.8975	0.9330	0.8645	1.0000	-0.4168
acceleration	0.4233	-0.5047	-0.5438	-0.6892	-0.4168	1.0000

Table 4: Correlation Matrix

4 Data visualization

We created a [pairplot](#) to represent each attribute value against each other in pairs. This has been crucial for getting an initial overview of the attribute values and identifying possible trends or correlations between them.

Many interesting (and sometimes expected) trends can be observed from the plot. For example, the cylinders vs. mpg plot shows that cars with more cylinders tend to consume more fuel. Similar patterns are seen with cars having higher displacement or greater weight. Other manufacturing-related trends include the fact that cars with more cylinders generally have higher (and more varied) displacement, horsepower, and weight, and take less time to accelerate from 0-60 mph. Some of these trends will be analyzed in further detail later. On the other hand, some correlations are less clear (or non-existent), particularly those related to acceleration, such as the relationship with mpg or weight.

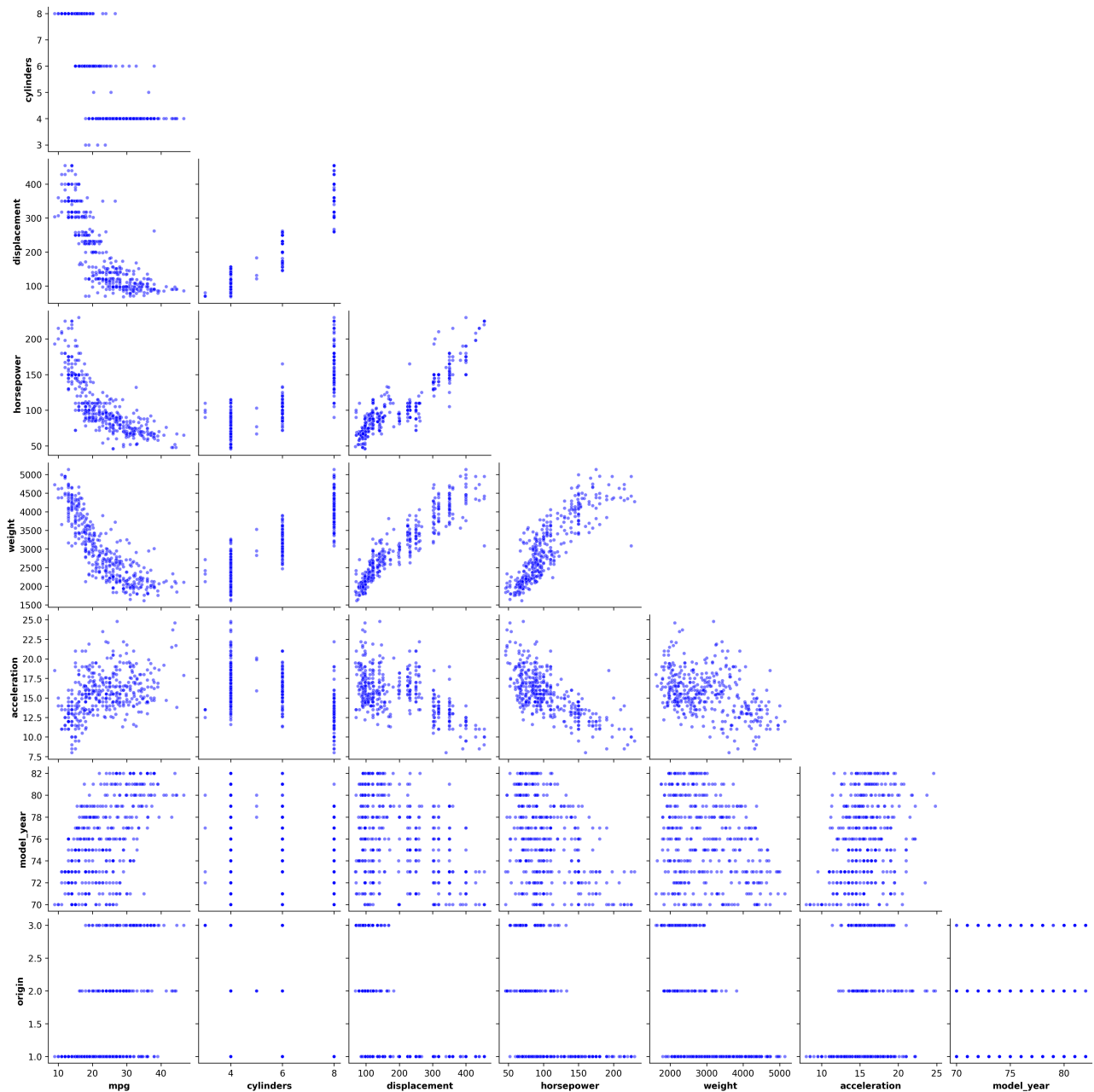


Figure 1: Pairplot showing pairwise relationships and correlation among the attributes of interest in the dataset. Only numerical attributes are displayed. Some degree of correlation is visible between almost all pairs of attributes.

Box plots are a suitable way to graphically display some of the data already presented in [summary statistics table](#) with the addition of a look at outliers, which will be analyzed in the [discussion section](#).

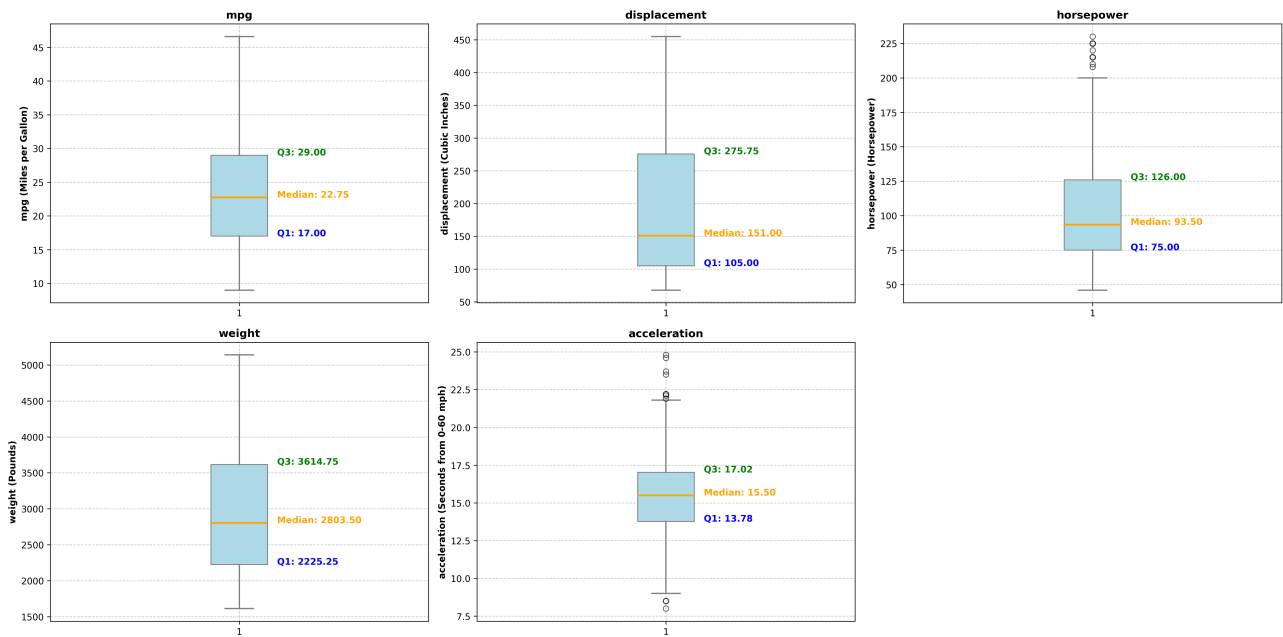


Figure 2: Box plots summarising the data set and displaying potential outliers values for each continuous attribute. In fact, *horsepower* and *acceleration* show a significant number of them.

The following histograms display the distribution of values for each continuous attribute. At first glance, only acceleration appears to follow a near-normal distribution, while the other attributes exhibit skewed distributions.

Attribute Data Distributions

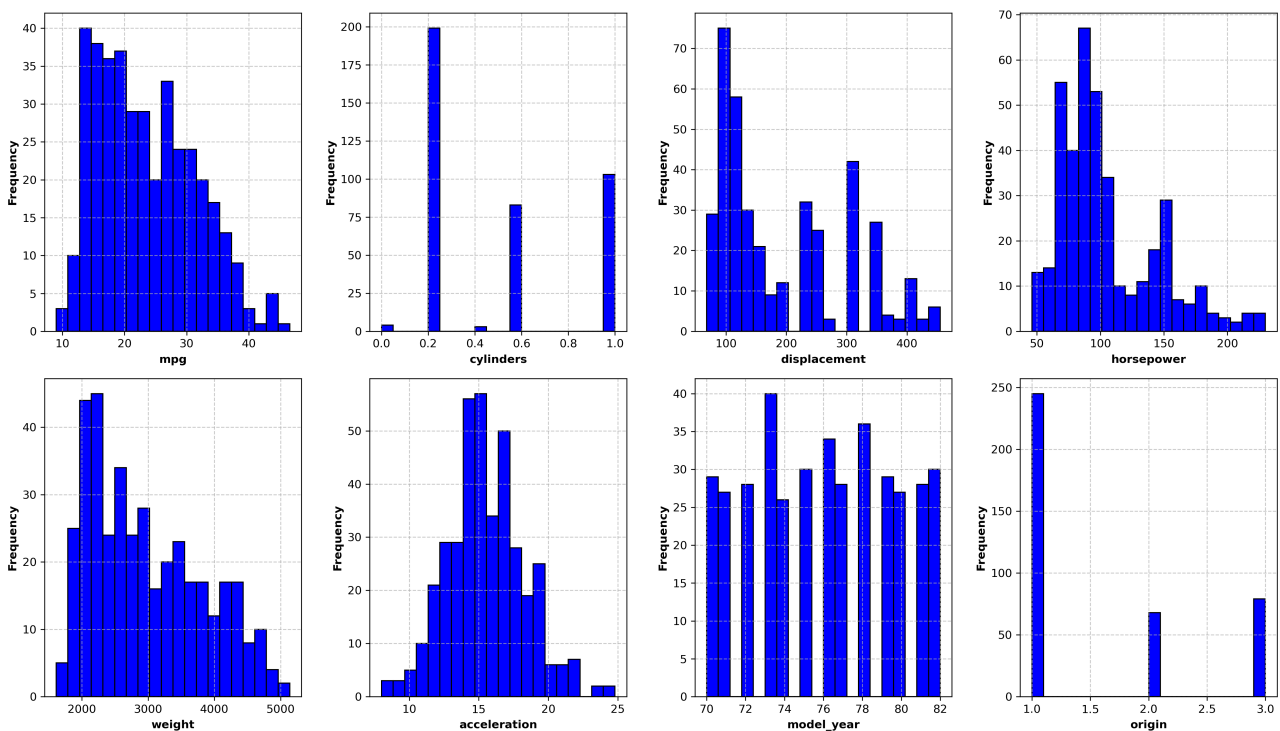


Figure 3: Histograms displaying the data distribution of all continuous attributes (plus standardized cylinders).

For the feasibility of our [classification task objective](#), we also examined one of the attributes that, by intuition, might influence the *origin* (target label). The following boxplot clearly shows that the least efficient cars are primarily observed in the USA, while Europe and Japan tend to prefer more fuel-efficient cars, possibly due to different regulations or fuel costs. This provides an initial indication that the dataset contains information that could ultimately be useful for predicting the continent in which a car may be sold.

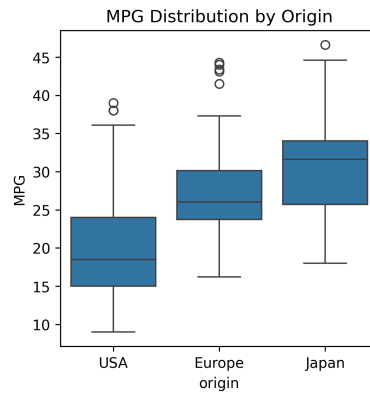


Figure 4: Boxplot of *mpg* by *origin*

Thinking about the [regression task objective](#), we decided to visualize the predictability of MPG based on some key attributes that intuitively suggest a car may be less efficient. We expected that heavier cars, with high displacement, high horsepower, and many cylinders, would be more performant and, therefore, consume more fuel. This will also be one of the main findings of the [PCA analysis](#).

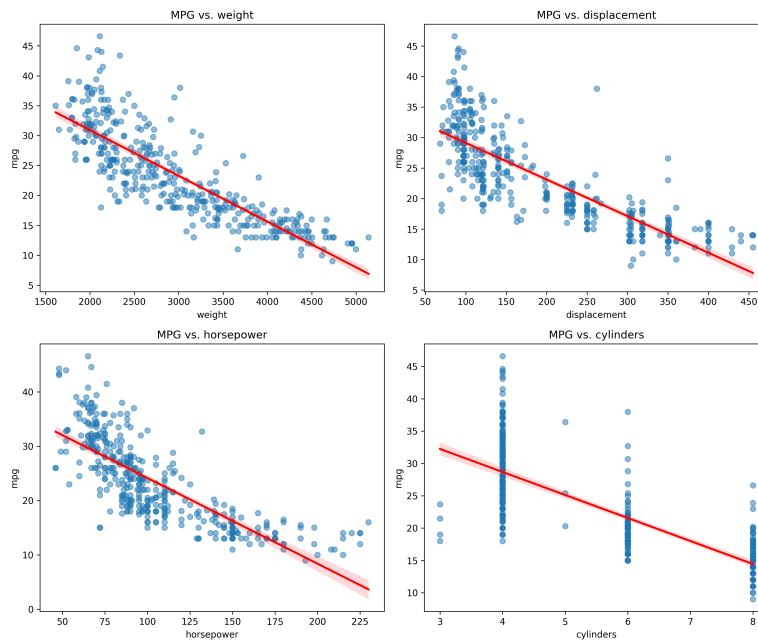


Figure 5: Comparison between MPG and some attributes that intuitively may influence it.

The heatmap below provides a more intuitive and visually appealing representation of the values obtained in the [correlation table](#), making it easier to identify patterns and relationships between variables at a glance.

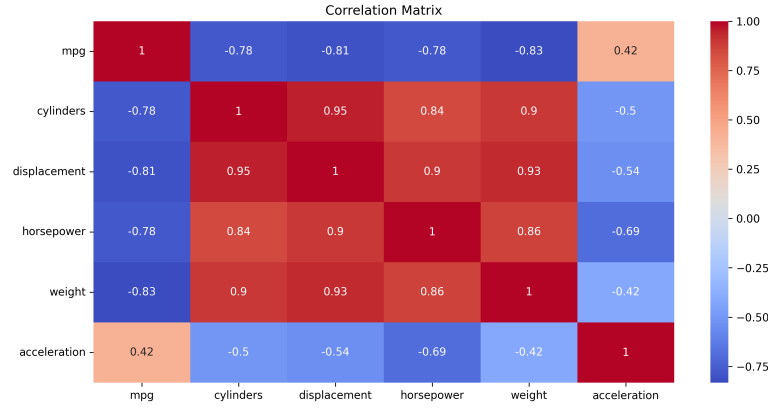


Figure 6: Correlation Matrix

5 Principal Component Analysis

Principal Component Analysis (PCA) is a powerful tool in Machine Learning that helps simplify complex datasets. When working with data, we often have many features (columns), and it's not always clear which ones are essential. Removing features without careful consideration can mean losing valuable information, while keeping everything might introduce noise or distortions due to differences in scale.

Some features might not contribute much to the overall representation of the data, while others may dominate just because of their magnitude, leading to biased results in downstream modeling. To address this, we apply Singular Value Decomposition to the standardized matrix:

$$\tilde{X} = U\Sigma V^T \quad (2)$$

Where:

- $\tilde{X}$ is the standardized data matrix of size $m \times p$, where m is the number of observations and p is the original number of features (see later).
- U is an orthogonal matrix of size $m \times m$, whose columns are the eigenvectors of $\tilde{X}\tilde{X}^T$. These represent the principal directions in the observation space.
- Σ is a diagonal matrix of size $m \times p$ containing the singular values in descending order, which indicate the contribution of each principal component to the retention of variance.
- V^T is the transpose of an orthogonal matrix V of size $p \times p$, whose columns are the eigenvectors of $\tilde{X}^T\tilde{X}$. These define the principal directions in the feature space.

As recommended, we included in $\tilde{X}$ only ratio attributes such as displacement, weight, horsepower, acceleration, and mpg, but we also decided to keep the (also standardized) discrete attribute *cylinders*. We believed it played a crucial role in representing the dataset accurately, ensuring we retained important structural information.

After obtaining the Σ and V matrices through the `dtumilttools` Python library, we can extract the first n principal components based on our threshold for cumulative explainability, which can be analyzed from the diagonal values of the Σ matrix.

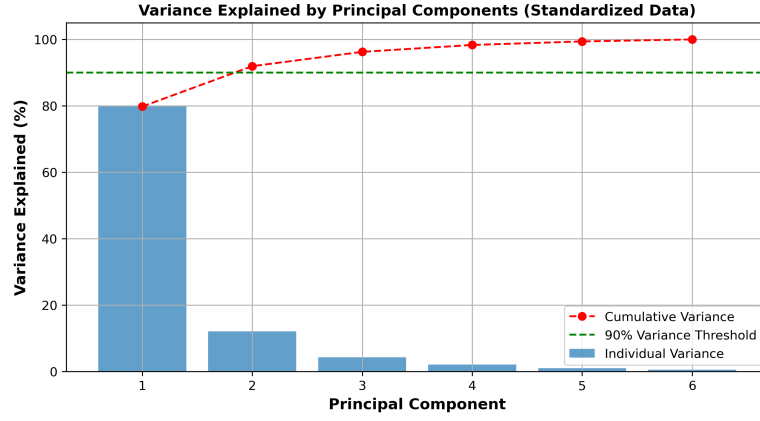


Figure 7: The bar plot represents the variance explained by each principal component (PC). The red dashed line displays the cumulative variance explained by each component. The first two principal components together account for more than 90% of the total variance, as indicated by the green threshold line. This justifies reducing the dataset to only two components, as they retain most of the original information while significantly reducing the dimensionality.

Attribute	PC1	PC2
MPG	0.39897	0.24483
Cylinders	-0.43062	-0.14831
Displacement	-0.44353	-0.10850
Horsepower	-0.43412	0.16616
Weight	-0.43010	-0.28610
Acceleration	0.29193	-0.89265

Table 5: Principal Component 1 (PC1) mainly separates heavy, high-displacement cars with more cylinders from lighter, fuel-efficient cars, while Principal Component 2 (PC2) captures differences in acceleration and weight, distinguishing fast-accelerating from slower-accelerating vehicles.

Once the n principal directions are selected, we project the entire dataset onto them using the following formula:

$$Z = \tilde{X}V_n \quad (3)$$

Where:

- Z is the transformed dataset of size $m \times n$, where each observation is now represented in the principal component space.
- $\tilde{X}$ is the standardized data matrix of size $m \times p$, where p is the original number of features.
- V_n is the matrix of the first n principal components, extracted from V , of size $p \times n$.

Thus, we obtain a new dataset of size $m \times n$, which serves as the most informative representation of the original data. This transformed dataset is now ready for further processing tailored to the intended tasks.

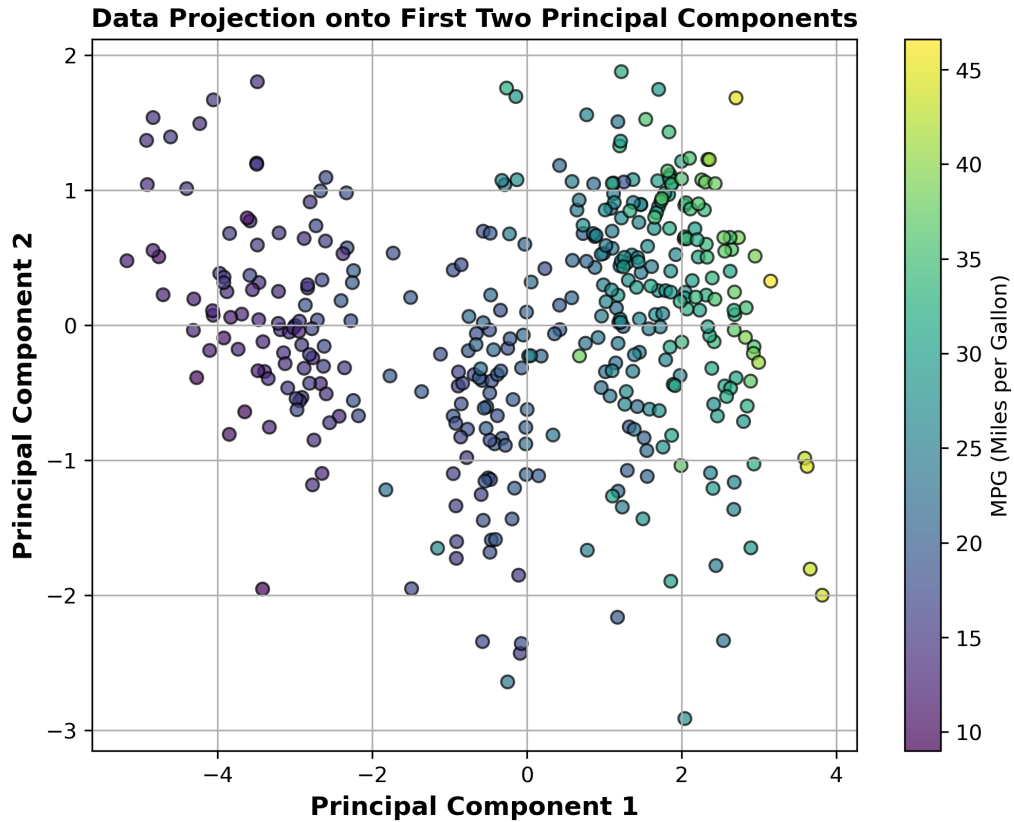


Figure 8: This scatter plot displays the dataset projected onto PC1 (horizontal axis) and PC2 (vertical axis), with a color-coding representing the *mpg* values (the target attribute of the classification task). PC1 (x-axis) mainly separates heavy, high-displacement cars on the left from lighter, fuel-efficient cars on the right, while PC2 (y-axis) captures differences in acceleration and weight, separating fast-accelerating cars at the top from slower-accelerating cars at the bottom. This projection shows how the relationship between fuel efficiency and vehicle characteristics is well preserved in the projected PCA data.

6 Discussion

In summary, this dataset contains 392 observations after filtering out records with missing values. These observations correspond to cars and include 8 features: *mpg*, *cylinders*, *displacement*, *horsepower*, *weight*, *acceleration*, *model year*, *origin*, and *car name*. It is important to note that the numerical values have been standardized to ensure consistency in the analysis.

Based on the data visualization and the [matrix correlation](#), it is observed that the features are highly correlated (as common sense might suggest), particularly *mpg*, with *displacement*, *horsepower*, and *weight*. A clear linear trend is visible among these attributes in [Figure 1](#), while the [correlation table](#) indicates that as characteristics such as *cylinders*, *displacement*, *horsepower*, and *weight* increase, fuel efficiency (*mpg*) tends to decrease. This strong and negative correlation is consistent with physical intuition, as heavier cars or those with larger engines generally consume more fuel.

Additionally, the [boxplots](#) indicate that outliers are present only in *horsepower* and *acceleration*. However, these values in the dataset are not problematic because they are not improbable in the context of car characteristics. In other words, these extreme values are possible for attributes like *horsepower* and *acceleration*. Furthermore, the *acceleration* and *horsepower* values for these outlier cars do not diverge much from the IQR and represent realistic engine specifications of more powerful vehicles. Therefore, they do not represent erroneous data points that need to be adjusted or removed.

Furthermore, Principal Component Analysis (PCA) reveals that the first two principal components together ac-

count for more than 90% of the total variance. This high level of explained variance suggests that the dataset’s dimensionality can be reduced to two components while preserving most of the original information. The projection of data onto these principal components also highlights a strong relationship between displacement, number of cylinders, acceleration, and weight in predicting a car’s fuel efficiency.

Considering these findings, we have a hint that the *mpg* regression may lead to noteworthy results. However, the classification of a car’s *origin* may present challenges, depending on how well the features differentiate the categories.

Overall, these results indicate that this dataset is suitable for achieving the regression and classification objectives.

7 About LLM usage

Although the LLM was used to enhance students’ understanding of various concepts, including data analysis and machine learning methodologies, it was not involved in the actual writing of this document, except for paraphrasing content originally written by hand.

References

- [1] R. Quinlan. *Auto MPG*. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5859H>. 1993.
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- [4] Tue Herlau, Mikkel N Schmidt, and Morten Mørup. “Introduction to machine learning and data mining”. In: *Lecture notes of the course of the same name given at DTU (Technical University of Denmark)* (2023).
- [5] Carnegie Mellon University. *Cars Dataset Description*. n.d. URL: <http://lib.stat.cmu.edu/datasets/cars.desc>.